INCH-POUND
MIL-DTL-32015A(AS)
5 May 2004
SUPERSEDING
MIL-DTL-32015
15 October 1999

DETAIL SPECIFICATION

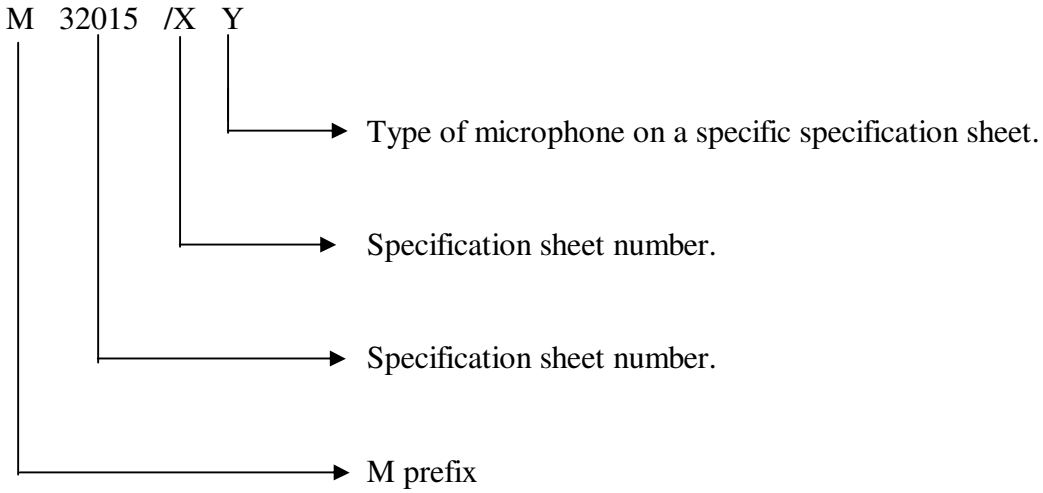
MICROPHONES, LINEAR, GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Air Systems Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers direct currant (dc) powered, high-gain, linear noise-cancelling microphones for boom and mask mounting applications.

1.2 Part or identifying number (PIN). PINs to be used for linear microphones acquired to this specification are created as follows:



Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division, Code 414100B120-3, Highway 547, Lakehurst, NJ, 08733-5100 or emailed to Tom.Omara@navy.mil. Since contact information can change you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil.

AMSC N/A FSC 5965

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in section 3 and 4 of this specification. This section does not include documents cited in other sections of this specification recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-55668	-	Cord, Electrical; Audio, Subminiature (Retractile and Straight)
MIL-DTL-32015/1	-	Microphones, Linear, 28v, Boom TYPE M-177/AIC
MIL-DTL-32015/2	-	Microphones, Linear, MBU-12-20P And, Oxygen Mask M-176/AIC And MBU-23/P
MIL-DTL-32015/4	-	Microphones, Linear, 10v, Oxygen Mask
MIL-DTL-32015/5	-	Microphones, Linear, 10v, Chemical, Biological, and Radiological (CBR)
MIL-DTL-32086	-	Intercom, Portable

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-464	-	Electromagnetic Environmental Effects Requirements for Systems
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests

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DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-217	-	Reliability Prediction of Electronic Equipment
MIL-HDBK-235/1	-	Electromagnetic (Radiated) Environment Considerations for Design and Procurement of Electrical and Electronic Equipment, Subsystems, and Systems Part 1B
MIL-HDBK-237	-	Electromagnetic Environmental Effects on Platforms, Systems, and Equipment

(Copies of these documents are available on line at http://assist.daps.dla.mil/quicksearch/ or www.dodssp.daps.mil or from the Standardization Document Order Desk, 700 Robins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents. drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS

ARMY

SM-B-883830	-	Microphone Boom

(A copy of the above drawing is available from the Commander, US Army-CECOM. Attn: AMSEL-IM-T-CS-A, Fort Monmouth, NJ 07703-5018.)

NAVAL AIR SYSTEMS COMMAND

NAWCADLKE-MISC-414100-0006	-	SK-N-864, Simulated Gun Blast Producing Equipment

(A copy of the above drawing is available from the Commander, Naval Air Warfare Center Aircraft Division, Code 7.2, Highway 547, Lakehurst, NJ 08733-5090.)

AIR FORCE

57B12662	-	Connector, Plug, Electrical U-173/U
57B12664	-	Tip-Electrical Plug Connector

(Copies of the above drawings are available from Defense Supply Center Columbus, DSCC-VTCD, P.O. Box 3990, Columbus, OH 43216-5000.)

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PUBLICATIONS

NAVAIR AD 1115- Electromagnetic Compatibility Design Guide for Avionics and Related Ground Support Equipment

(A copy of the above publication is available from the Commander, Naval Air Systems Command, Code AIR-7.2.5.3, Building 407, Naval Air Station, Patuxent River, MD 20670.)

CRDC-SP-84010 - United States Joint Services Standardization Agreement for Fit Factor Testing of Military Masks and ERDEC testing protocol entitled "ERDEC Respirator Quantitative Fit Testing Program"

(A copy of the above publication is available from the Commander, Naval Air Systems Command, Code AIR-4.6.3.1, Building 2187, Naval Air Station, Patuxent River, MD 20670.)

NAVSEA TE-00-AB-GTP-010 – Part Stress Derating

(A copy of the above manual is available from Naval Inventory Control Point, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094, or email www.navicp.navy.mil)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI-S3.2 - Speech Over Communication Systems, Method of Measuring the Intelligibility of

(A copy of this document is available from the American National Standards Institute, 25 West 43rd Street, 4th floor, New York, NY 10036, or email http://www.ansi.org/)

THE INSTITUTE FOR INTERCONNECTING AND PACKAGING ELECTRONIC CIRCUITS (IPC, INC.)

J-STD-001 - Requirements for Soldered Electrical and Electronic Assemblies

(A copy of this document is available from the Institute for Interconnecting and Packaging Electronic Circuits, 2215 Sanders Road, Northbrook, IL 60062-6135, or email http://www.ipc.org/)

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2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.3 Component parts. The microphone shall be supplied with component parts as specified in the applicable specification sheet (see 3.1).

3.4 Weight. The weight of the microphone shall be as specified in the specification sheet.

3.5 Color. The exterior surface of the microphone assembly shall be as specified in the specification sheet.

3.6 Design. The design of the microphone shall meet the mechanical construction (see 3.7), electrical characteristics (see 3.8.1), and acoustical performance requirements (see 3.8.2) of this specification.

3.6.1 Aging. The microphone shall not contain parts fabricated from materials known to change their properties because of aging.

3.6.2 Materials. Materials shall be as specified herein. However, when a definite material is not specified, a material shall be used that shall enable the microphones to meet the performance requirements of this specification. Dissimilar metals shall not be used in direct contact unless specifically required. Metals shall be protected against the effects of electrolytic action. Acceptance or approval of any constituent material shall not be construed as a guarantee of the acceptance of the finished product. When specified in the specification sheet, materials shall be resistant, or suitably treated, to resist exposure to chemical and biological agents.

3.7 Construction.

3.7.1 Unacceptable material. Natural rubber or natural rubber cement shall not be used in the microphone. Rubber substitutes and plastic materials shall not be used as edges and fillers.

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3.7.2 Microphone material. The voltage-generating element shall be an electret condenser, a piezoelectric ceramic, a polyvinylidene fluoride (PVF2), or an equivalent material giving demonstrated similar results. Microphone designs incorporating magnetic material shall not be used.

3.7.3 Microphone amplifier. The microphone shall be designed with an amplifier capable of accepting impressed power from the intercommunication systems.

3.7.4 Cement. To ensure uniform adhesion and strength characteristics, any cement used shall be properly and uniformly controlled and cured throughout production.

3.7.5 Microphone shield. To preclude the effects of electromagnetic interference or susceptibility, the microphone transducer and amplifier shall be shielded.

3.7.6 Microphone connector. The microphone connector shall be as specified in the applicable specification sheet.

3.7.7 Microphone positioning adjustment. The microphone positioning adjustment for microphones incorporating a swivel type mounting mechanism shall provide a noncritical adjustment to obtain a torque of 30 to 50 inch-ounces. From an initial setting of 40 inch-ounces, the torque shall not change in excess of 20 percent after subjection to any or all of the environmental tests. The swivel shall be designed such that it may be adjusted over full range of swivel for at least one hundred times before it must be readjusted for specified torque. Total operating life of the swivel mechanism shall be at least ten thousand swivels over full range at specified torque.

3.7.8 Solderability. Soldering shall be in accordance with J-STD-001.

3.7.9. Deleted

3.8 Performance.

3.8.1 Electrical characteristics.

3.8.1.1 Input current. The DC input current shall be as specified in the applicable specification sheet.

3.8.1.2 Output impedance. The electrical output impedance of the microphone shall be as specified in the applicable specification sheet.

3.8.1.3 Insulation resistance. The insulation resistance between the conductors and case shall be not less than 10 megohms.

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3.8.1.4 Dielectric withstanding voltage. The microphone shall withstand the application of a 115 volt nominal alternating potential of commercial line frequency without flashover or breakdown when tested as specified (see 4.5.2.4).

3.8.1.5 Voltage operating range. The voltage operating range shall be as specified in the applicable specification sheet.

3.8.2 Acoustical.

3.8.2.1 Sensitivity.

3.8.2.1.1 Sensitivity to a close sound source. The sensitivity to a close sound source shall be as specified in the applicable specification sheet. After being subjected to any or all environmental tests, sensitivity shall not vary by more than ?3 dB from the initial reading.

3.8.2.1.2 Sensitivity at reduced bias voltage. At DC operating voltages below the normal range, the output of the microphone shall be as specified in the applicable specification sheet.

3.8.2.1.3 Sensitivity at altitude. The sensitivity of the microphone at a simulated altitude of 15,000 feet shall be as specified in the applicable specification sheet.

3.8.2.2 Frequency response.

3.8.2.2.1 Frequency response to a close sound source. The frequency response characteristic of the microphone assembly to a close sound source shall be as specified in the applicable specification sheet. After subjection to any or all environmental tests, the frequency response shall not vary by more than ?3 dB from the initial readings.

3.8.2.2.2 Frequency response at altitude. The frequency response characteristics of the microphone at a simulated altitude of 25,000 feet shall be as specified in the applicable specification sheet.

3.8.2.3 Intelligibility. The microphones shall be designed to enable the listeners to achieve a score of 90 percent or better when tested with one of the sets of speech materials specified in ANSI S3.2.

3.8.2.4 Noise-immunity characteristic. The noise-immunity characteristic of the microphone shall be as specified in the applicable specification sheet.

3.8.2.5 Linearity. The output of the microphone shall be a linear function of the sound pressure level (SPL) input to the microphone in the range 74 to 124 dB SPL or as specified in the applicable specification sheet.

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3.8.2.6 Harmonic distortion. The harmonic distortion in the output of the microphone shall be no greater than 2 percent up to 114 dB SPL and no greater than 5 percent from 114 to 124 dB SPL or as specified in the applicable specification sheet.

3.8.2.7 Gain stabilization time when specified. The gain stabilization time shall be as specified in the applicable specification sheet.

3.8.2.8 Over voltage protection. Unless otherwise specified the microphone shall withstand overvoltage conditions as specified in the applicable specification sheet. Upon removal of the overvoltage condition the microphone shall resume normal operation when the voltage is returned to within standard operating limits. The microphone shall meet the sensitivity to a close sound source within ? 3 dB from the initial reading.

3.8.3 Cord, cordage, cables, connectors, and pins.

3.8.3.1 Temperature range. The cable shall be flexible and resilient throughout the temperature range of -40 to +85 ?C and shall show no evidence of cracking or any other damage as a result of exposure to the temperature range specified herein.

3.8.3.2 Flex life. The unterminated cable shall have a mean flex life of 25,000 flexes without showing evidence of the following:

a. Damage to the outside jacket (splits, cracks)
b. Damage to the insulation of the individual conductors
c. Electrical discontinuity

3.8.3.3 Cord anchorage. Unless specified in the specification sheet, the cord and its anchorage in the two connector plugs shall withstand an 8 pound pull applied to the connector plug for a period of 30 minutes without damage to the cord and anchorage. Any slippage at the connector shall be considered a failure.

3.8.3.4 Conductor strain relief and twist relief. The conductors shall be anchored in the two-connector plugs so as to provide strain relief for the conductors. Means shall be provided to prevent twisting or inward movement of the cord in the connector plug.

3.8.3.5 Dielectric withstanding voltage of cord assembly. For the finished cord assembly, the insulation between conductors shall withstand the application of a 500 volt alternating potential of commercial line frequency for 5 seconds without flashover or breakdown.

3.8.3.6 Insulation resistance of cord assembly. The insulation resistance of the finished cord assembly shall be 10 meg-ohms minimum.

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3.8.3.7 Pin retention. All connector, microphone and amplifier pins shall withstand a pull force of 20 ± 2 pounds for 30 seconds without slippage or damage. If the connector is molded, the applied force shall be 30 ± 2 pounds.

3.8.3.8 Amplifier/microphone mounting screw torque/crush torque. The screw mounting holes shall be constructed in such a manner that the amplifier shall not be damaged when the mounting screws are torqued to 3 inch pounds.

3.8.4 Environmental.

3.8.4.1 Temperature-altitude. The microphone assembly shall meet the requirements of 3.8.2.1.1 and 3.8.2.2.2 during periods of operation as detailed in the temperature-altitude test. (see 4.5.5.1).

3.8.4.2 Humidity. Upon completion of the humidity test (see 4.5.5.2), the microphone assembly shall meet the requirements of 3.8.2.1.1 and 3.8.2.2.1.

3.8.4.2.1 Humidity condensing in the mask. Upon completion of the humidity condensing in the mask test (see 4.5.5.14) the microphone assembly shall meet the requirements of 3.8.2.1.1 and 3.8.2.2.1.

3.8.4.3 Immersion. Upon completion of the immersion test (see 4.5.5.3), the microphone assembly shall meet the requirements of 3.8.2.l.1 and 3.8.2.2.1.

3.8.4.4 Dust. Upon completion of the dust test (see 4.5.5.4), the microphone assembly shall meet the requirements of 3.8.2.1.1 and 3.8.2.2.1.

3.8.4.5 Blast. Upon completion of the blast test (see 4.5.5.5), the microphone assembly shall meet the requirements of 3.8.2.1.1 and 3.8.2.2.1.

3.8.4.6 Vibration. Upon completion of the vibration test (see 4.5.5.6), the microphone assembly shall meet the requirements of 3.8.2.1.1 and 3.8.2 2.1.

3.8.4.7 Shock-drop. Upon completion of the shock-drop test (see 4.5.5.7), the microphone assembly shall meet the requirements of 3.8.2.1.1. and 3.8.2.2.1. There shall be no evidence of breaking, cracking, or other structural damage.

3.8.4.8 Electromagnetic environmental effects (E3) requirements. The microphone assembly shall be designed, analyzed, installed, and tested to perform its mission in the worldwide electromagnetic environment (EME) without any malfunction or degradation of performance.

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3.8.4.8.1 Electromagnetic compatibility (EMC). The microphone assembly shall demonstrate electromagnetic compatibility (EMC). The microphone assembly shall perform its individually designed functions in a common EME without causing or suffering unacceptable degradation due to electromagnetic interference (EMI) to or from other equipment/systems in the same environment in accordance with the requirements of MIL-STD-464. EMC managerial guidelines are provided by MIL-HDBK-237.

3.8.4.8.2 Electromagnetic interference (EMI). The generation of EMI by the microphone assembly and the susceptibility of the microphone assembly to EMI shall be in accordance with the design limits of MIL-STD-461 and as modified by this specification. These requirements shall be met in all microphone assembly operating modes.

3.8.4.8.2.1 Electromagnetic emissions. Not applicable.

3.8.4.8.2.2 Electromagnetic susceptibility. The fully hardened microphone assembly shall be capable of surviving exposure to the expected radiated EME in accordance with MIL-HDBK-235 and Table IA of MIL-STD-464 External EME for systems capable of shipboard operations. Radiated susceptibility shall be conducted in accordance with 4.5.5.8.3.2.

3.8.4.8.2.3 EMI susceptibility. The following criteria shall be applied to determine whether the microphone assembly is susceptible. The audio output of the amplifier shall be monitored by use of an appropriate signal analyzer. The interference-induced distortion in the audio output signal shall be not greater than 12 percent.

3.8.4.9 Explosive atmosphere. When specified in the specification sheet, the microphone assembly shall be subjected to the explosive atmosphere test of MIL-STD-810 Test Method 511.3, Procedure I. In addition to meeting the requirements of MIL-STD-810, the microphone shall not show evidence of damage from exposure to the explosive atmosphere environment. Upon completion of the test the microphone shall meet the requirements of 3.8.2.1.1 and 3.8.2.2.1.

3.8.4.10 Environmental stress screening (ESS). When specified in the specification sheet (see 3.l) the contractor shall subject each microphone assembly to, and successfully pass ESS at the completion of quality conformance inspection and prior to delivery.

3.8.4.11 Salt fog. Upon completion of the salt fog test (see 4.5.5.11), the microphone shall show no corrosion of finishes and metals. Such corrosion shall be defined as any visible degradation of the equipment surfaces that can be attributed to flaky, pitted, blistered, or otherwise loosened finish of the metal surface.

3.8.4.12 Fungus. Only fungus inert materials shall be used in the fabrication of the microphone assembly.

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3.8.4.13 Mask pressurization. When specified in the specification sheet, the microphone assembly shall be assembled in a mask and tested in accordance with 4.5.5.13. Upon completion of the test, the microphone assembly shall meet the requirements of 3.8.2.1.1 and 3.8.2.2.1.

3.8.4.14. Condensing Humidity. When specified in the specification sheet, the microphone assembly shall be subjected to the condensing humidity test (see 4.5.5.14).

3.8.4.15. Rapid decompression. When specified in the applicable specification sheet, the microphone shall be tested in accordance with the low-pressure altitude test of MIL-STD-810, Method 500.4, Procedure III, and 4.5.5.15. The microphone shall withstand a simulated loss of cabin pressure from 8,000 to 40,000 feet in not more than 15 seconds. The microphone assembly shall fully recover without damage or degradation within 2.0 seconds following the decompression and meet the requirements of 3.8.2.1.1, 3.8.2.2.1, and 3.8.2.6.

3.8.4.16. Explosive decompression. When specified in the applicable specification sheet, the microphone shall be tested in accordance with the low-pressure altitude test of MIL-STD-810, Method 500.4, Procedure IV and 4.5.5.16. The microphone shall withstand a simulated loss of cabin pressure from 8,000 to 23,000 feet in 0.1 second and from 20,500 to 50,000 feet in 0.1 second. The microphone assembly shall fully recover without damage or degradation within 1.0 second following the decompression and meet the requirements of 3.8.2.1.1, 3.8.2.2.1, and 3.8.2.6.

3.8.4.17 Rain. When specified in the specification sheet, the microphone assembly shall be tested in accordance with 4.5.5.17 and shall meet the requirements of 3.8.2.1.1 and 3.8.2.2.1

3.8.4.18 Chemical agent permeation. When specified in the specification sheet, the microphone assembly shall meet the requirements below.

3.8.4.18.1 Mustard. The microphone assembly shall be subjected to the tests of 4.5.5.18.1 without any degradation of performance.

3.8.4.18.1 Soman. The microphone assembly shall be subjected to the tests of 4.5.5.18.2 without any degradation of performance.

3.8.4.19 Quantitative fit factor (QFF). When specified in the specification sheet, the microphone assembly shall meet the requirements below when tested in accordance with 4.5.5.19.

3.8.4.19.1 QFF normal mode (chemical protection). For chemical protection, the QFF, with the ventilating flow "ON" shall be not less than $2x10^4$ fit factor (FF) when tested as specified in 4.5.5.19.1. The protection factor shall be achieved continuously for not less than 16 hours.

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3.8.4.19.2 QFF Failure mode. For chemical protection, the QFF with the ventilating flow "OFF" shall not be less than $1x10^3$ (FF). when tested as specified in 4.5.5.19.2 The protection factor shall be achieved continuously for not less than 16 hours

3.9 Identification of product. The identification marking on the microphone assembly shall include the information as specified in the specification sheet.

3.10 Reliability. The microphone assembly shall meet a reliability minimum of ?o = 9,000 hours with a duty cycle of one minute on to 12 seconds off.

3.11 Workmanship. Microphone assemblies shall be processed in such a manner as to be uniform in quality and shall be free from defects that will affect life, functioning, and appearance. After fabrication, parts and the assembled microphone shall be cleaned of smudges; loose, spattered, or excess solder; weld metal; and metal chips and mold release agents; or any other foreign material that might detract from the intended operation, function, or appearance of the microphone. This shall include any particles that could loosen or become dislodged during the normal expected life of the microphone. All corrosive material shall be removed. Whenever possible, cleaning shall take place before the parts are assembled into the equipment. All assembled equipment shall be cleaned of contaminants such as lubricating oils, mold release agents, waxes, sand, corrosion products, solder fluxes, fingerprints, dust. The nature of the contaminant must be determined to the extent that a suitable cleaning solvent can be selected for item removal. The solvent must not damage the construction materials. Cleaning processes shall have no deleterious effect on the equipment or parts. There shall be no evidence of loose contacts; poor or improper molding or fabrication, damage or improperly assembled contacts; peeling, flaking or chipping of plating or finish; mechanical damage due to testing environments; nicks or burrs on surfaces of metal parts, improper or incorrect marking; or improper tinning of solder cups, terminal pins, or contacts.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in section 4 of this specification.

4.2.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities with the accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained by the contractor.

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4.2.2 Test fixtures. The fixtures that are used to position the calibrating microphone assembly or test microphone element with respect to the sound source shall be designed to minimize any possible error in acoustical measurements owing to vibration, sound transmitted from the artificial voice, unstable positioning of microphone elements, reflections from fixtures or environment such as the interior of an altitude chamber, and ambient noise. No baffles shall be near the sound-sensitive parts of the microphone elements.

4.2.3 Test conditions. Unless otherwise specified (see 6.2), the microphone shall be tested under the following conditions.

a. Temperature: Room ambient. (+l5 to +35 ?C)
b. Pressure: Normal atmosphere
c. Humidity: Prevailing ambient up to 90 percent relative humidity

4.2.4 Close sound source.

4.2.4.1 Simulator microphone signal (artificial voice). An artificial voice, such as a Bruel & Kjaer (B&K) type 4219, shall be used as the sound source in performing the sensitivity and frequency response to a sound-source characteristic measurement specified herein. The artificial voice shall be calibrated with a B&K type 4133 half-inch microphone, or demonstrated equivalent, which shall have calibration traceable to the National Institute of Standard Technology (NIST), formerly the National Bureau of Standards (NBS).

4.2.4.2 Reference sound pressure level. All measurements shall be made at a sound pressure level that shall allow the harmonic distortion (the noise produced in the test microphone by the air pumped through the orifice of the artificial voice at low frequencies) to be sufficiently small so as not to introduce a significant measurement error (less than ?1 percent). For the purpose of this specification, all sensitivity data, regardless of the actual sound pressure level used in the measurements, shall be referred to a sound pressure level of 104 dB relative to 20 ?Pa (micropascals).

4.2.4.3 Calibration of close sound source. The sound source shall be calibrated to produce the desired sound pressure at the grid of the B&K type 4133 microphone over the frequency range of 400 to 6,000 Hz. The calibration of the microphone will have been previously measured using a B&K type 4220 piston-phone or equivalent. The grid of the type 4133 microphone shall be spaced 0.25 inch from, and coaxial with, the mouth of the artificial voice during calibration. The sound source shall be calibrated at each altitude at which it is used.

4.2.5 Plotting of frequency response characteristic. The frequency response characteristic of the microphone, if plotted, shall be plotted on a graphic level recorder using semilogarithmic coordinate paper with the response in dB on the linear ordinate scale and the frequency in Hz on the logarithmic abscissa scale. The length of a 100 to 1 frequency interval on the abscissa scale shall equal the length of 50 dB on the ordinate scale.

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4. 3 First article inspection. The contractor shall perform first article inspection after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units, which have been produced with equipment and procedures normally used in production.

4. 3.1 Sample size. Ten microphones shall be subjected to first article inspection. Additionally, 5 unterminated cables shall be provided for flexing life testing (see 3.8.3.2).

4.3.2 Inspection routine. The sample shall be subjected to the inspections specified in table I, in the order shown. All sample units shall be subjected to the inspections of group I. The samples shall then be divided equally into two groups and subjected to the inspections for their particular group (for groups II and III).

4.3.3 Failures. More than one failure in group I and any failure in group II or III shall be causes for refusal to grant first article approval.

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TABLE I. First article inspection.

Inspection	Requirement paragraph	Test method paragraph
Group I (10 samples)		
Visual and mechanical inspection	3.3, 3.4, 3.5, 3.6, 3.7, 3.9, and 3.11	4.5.1
Impedance*	3.8.1.2	4.5.2.2
Insulation resistance	3.8. 1.3	4.5.2.3
Dielectric withstanding voltage	3.8.1.4	4.5.2.4
Sensitivity to a close sound source	3.8.2.1.1	4.5.3.1.1
Sensitivity at reduced bias voltage	3.8.2.1.2	4.5.3.1.2
Sensitivity at altitude *	3.8.2.1.3	4.5.3.1.3
Frequency response to a close sound source	3.8.2.2.1	4.5.3.2.1
Frequency response at altitude	3.8.2.2.2	4.5.3.2.2
Group II (5 samples)		
Intelligibility	3.8.2.3	4.5.3.3
Noise immunity	3.8.2.4	4.5.3.4
Linearity	3.8.2.5	4.5.3.5
Harmonic distortion *	3.8.2.6	4.5.3.6
Gain stabilization time	3.8.2.7	4.5.3.7
Overvoltage protection	3.8.2.8	4.5.3.8
Cord, cordage, and cable		
Temperature range	3.8.3.1	4.5.4.1
Flexing life (cable unterminated)	3.8.3.2	4.5.4.2
Cord anchorage	3.8.3.3	4.5.4.3

* Indicates minimum tests for first article testing (see 4.3).

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TABLE I. First article inspection. - (Continued).

Inspection	Requirement paragraph	Test method paragraph
Conductor strain relief and twist relief	3.8.3.4	4.5.4.4
Dielectric withstanding voltage of cord assembly	3.8.3.5	4.5.4.5
Insulation resistance of cord assembly	3.8.3 6	4.5.4.6
Pin retention	3.8.3.7	4.5.4.7
Mounting screw torque	3.8.3.8	4.5.4.8
Group III (5 samples) Environmental Subgroup A (2 samples)		
Temperature-altitude *	3.8.4.1	4.5.5.1
Vibration	3.8.4.6	4.5.5.6
Humidity *	3.8.4.2	4.5.5.2
Blast	3.8.4.5	4.5.5.5
Subgroup B (2 samples)		
Electromagnetic environmental effects (E3)*	3.8.4.8	4.5.5.8
Explosive Atmosphere	3.8.4.9	4.5.5.9
Salt fog *	3.8.4.11	4.5.5.11
Subgroup C (1 sample)		
Dust	3.8.4.4	4.5.5.4
Shock-drop *	3.8.4.7	4.5.5.7
Immersion	3.8.4.3	4.5.5.3
Fungus	3.8.4.12	4.5.5.12
Mask Pressurization*	3.8.4.13	4.5.5.13
Condensing Humidity*	3.8.4.14	4.5.5.14
Rapid decompression *	3.8.4.15	4.5.5.15
Explosive decompression *	3.8.4.16	4.5.5.16
Rain*	3.8.4.17	4.5.5.17
Chemical Agent Permeation*	3.8.4.18	4.5.5.18
Mustard*	3.8.4.18.1	4.5.5.18.1
Soman*	3.8.4.18.2	4.5.5.18.2
Quantitative Fit Factor (QFF)*	3.8.4.19	4.5.5.19
QFF Normal mode(Chemical)*	3.8.4.19.1	4.5.5.19.1
QFF Failure Mode*	3.8.4.19.2	4.5.5.19.2

* Indicates minimum tests for first article testing (see 4.3).

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4.4 Conformance inspection.

4.4.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections. Except as specified in 4.4.1.4.4, delivery of products, which have passed groups A and B inspections shall not be delayed pending the results of group C inspection.

4.4.1.1 Inspection lot. An inspection lot shall consist of all microphones of the same type, produced under the same conditions, and offered for inspection at one time.

4.4.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table II, in the order shown.

4.4.1.2.1 Sampling plan. A sample of parts shall be randomly selected in accordance with table III. If one or more defects are found, the lot shall be rescreened and defects removed. After screening and removal of defects, a new sample of parts shall be randomly selected in accordance with table III, if one or more defects are found in the second sample, the lot shall be rejected and shall not be supplied to this specification.

TABLE II. Group A inspection.

Group A Inspection	Requirement paragraph	Test method paragraph
Visual and mechanical inspection	3.3, 3.4, 3.5, 3.6, 3.7, 3.9, and 3.11	4.5.1
Impedance	3.8.1.2	4.5.2.2
Sensitivity to a close sound source	3.8.2.1.1	4.5.3.1.1

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TABLE III. Group A zero defect-sampling plan.

LOT SIZE	SAMPLE SIZE		
	ELECTRICAL CHARACTERISTICS	VISUAL and MECHANICAL EXAMINATION	
		MAJOR	MINOR
2 to 8	100 percent	100 percent	3
9 to 15	100 percent	13	3
16 to 25	100 percent	13	3
26 to 50	100 percent	13	5
51 to 90	100 percent	13	6
91 to 150	125	13	7
151 to 280	125	13	10
281 to 500	125	29	11
501 to 1,200	125	34	15
1,201 to 3,200	125	42	18
3,201 to 10,000	294	50	22
10,001 to 35,000	294	60	29

Note: If sample size equals or exceeds lot size, 100 percent inspection is required.

4.4.1.3 Group B inspection. Group B inspection shall consist of the tests specified in table IV and shall be made on sample units that have been subjected to and passed group A inspection.

4.4.1.3.1 Sampling plan. A sample of parts shall be randomly selected in accordance with table V. If one or more defects are found, the lot shall be rescreened and defects removed. After screening and removal of defects, a new sample of parts shall be randomly selected in accordance with table V. If one or more defects are found in the second sample, the lot shall be rejected and shall not be supplied to this specification.

4.4.1.3.2 Disposition of samples. Sample units that have passed the group B inspection may be delivered on the contract or purchase order if the lot is accepted and the sample units are still within specified electrical tolerances.

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TABLE IV. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph
Frequency response to a close sound source	3.8.2.2.1	4.5.3.2.1
Noise immunity	3.8.2.4	4.5.3.4
Linearity	3.8.2.5	4.5.3.5
Harmonic distortion	3.8.2.6	4.5.3.6
Temperature range (cable)	3.8.3.1	4.5.4.1
Flex life (unterminated cable)	3.8.3.2	4.5.4.2
Conductor strain relief and twist relief	3.8.3.4	4.5.4.4

TABLE V. Group B zero defect sampling plan.

LOT SIZE	SAMPLE SIZE
2 to 8	2
9 to 15	2
16 to 25	3
26 to 50	5
51 to 90	6
91 to 150	7
151 to 280	9
281 to 500	11
501 to 1,200	13
1,201 to 3,200	15
3,201 to 10,000	15
10,001 to 35,000	15

NOTE: If sample size equals or exceeds lot size, 100 percent inspection is required.

4.4.1.4 Group C inspection. Group C inspection shall consist of the tests specified in table VI in the order shown. Group C inspection shall be made on sample units that have passed groups A and B inspections.

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TABLE VI. Group C inspection.

Inspection	Requirement paragraph	Test method paragraph
Subgroup 1 (5 sample units)		
Sub-Subgroup A (2 samples)		
Humidity	3.8.4.2	4.5.5.2
Sub-Subgroup B (2 samples)		
Sensitivity at altitude	3.8.2.1.3	4.5.3.1.3
Frequency response at altitude	3.8.2.2.2	4.5.3.2.2
Temperature-altitude	3.8.4. 1	4.5.5.1
Vibration	3.8.4.6	4.5.5.6
Immersion	3.8.4.3	4.5.5.3
Sub-Subgroup C (1 sample)		
Salt fog	3.8.4.11	4.5.5.11
Shock-drop	3.8.4.7	4.5.5.7
Subgroup 2(3 sample units) (1 unit per each test)		
Intelligibility	3.8.2.3	4.5.3.3
Dust	3.8.4.4	4.5.5.4
Blast	3.8.4.5	4.5.5.5

4.4.1.4.1 Sampling plan.

4.4.1.4.1.1 Sample size. Eight samples of the complete microphone assembly shall be selected at random; five samples shall be assigned for subgroup l and three samples for subgroup 2 of group C in table VI. The samples shall be selected at the start of the contract from the first quality conformance inspection lot. These samples shall constitute the group C requirement for the first 1,000 units produced. Thereafter, five samples of the complete assemblies shall be selected at random for subgroup l of group C in table VI. These samples shall be selected once each month, or every 1000 units, whichever occurs first.

4.4.1.4.2 Failures. If one or more sample units fail to pass group inspection, the lot shall be considered to have failed.

4.4.1.4.3 Disposition of sample. Sample units that have been subjected to group C inspection shall not be delivered on the contract or purchase order.

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4.4.1.4.4 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall notify the contracting activity and the cognizant inspection activity of such failure. The manufacturer shall then take corrective action on the materials or processes (or both) as warranted. The manufacturer shall also take corrective action on all units of product that can be corrected, that were manufactured essentially with the same materials and the same processes, and that are subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action acceptable to the qualifying activity has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (on all inspections or on the inspection that the original sample failed, at the option of the procuring activity). Groups A and B inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity.

4.4.1.4.5 Defects in items already accepted. The investigation of a test failure could indicate that defects might exist in items already accepted. If so, the contractor shall fully advise the contracting activity of all defects likely to be found and the method of correcting them.

4.5 Methods of inspection.

4.5.1 Visual and mechanical inspection. The microphone shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.3, 3.4, 3.5, 3.6, 3.7, 3.9, and 3.11). The defects shall be classified as specified in table VII. Particular attention shall be paid to the following where applicable.

a. Soldering operations
b. Uniformity and strength of cementing
c. Security and cleanliness of microphone terminal connections to cable terminals
d. Construction of microphone cord assembly
e. Vent in microphone case for pressure equalization
f. Cleanliness of vent holes throughout the microphone structure
g. Workmanship regarding the fit and sealing of the microphone case
h. Set screws holding the microphone plug to the microphone (slotted, headless screws shall be used)
i. Weight.

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TABLE VII. Classification of defects for visual and mechanical inspection.

Defect Type	Classification	
	Major	Minor
Dimensions Materials and finish	Dimensions not as specified. Materials not as specified. Wrong or incomplete finish. Large amounts of flaking, peeling, or chipping of finish.	Scratches, cuts, abrasions, and similar defects, causing exposure of base metal.
Parts	Missing parts. Inoperative, improperly assembled, or defective parts that could cause the unit to fail in service. Wrong parts.	Defective parts that would reduce efficiency of use, but not cause failure in service. Cracks or chipped surfaces not having an effect on the functioning, assembly, maintenance, or life of the unit.
Marking	Marking missing, illegible, or incorrect.	Marking dirty or smudged but legible
Foreign objects	Any metallic foreign object not firmly attached that could cause a short circuit or acoustical malfunctioning of the unit. Any nonmetallic foreign object such as insulation, dirt, or phenolic chips that could cause acoustical malfunctioning of the unit.	Any metallic or nonmetallic foreign object that affects appearance but could not cause acoustical malfunctioning of the unit.
Soldering	Improper wrap- Less than l/2 turn. Unsoldered joint- Solder not applied where intended. Insufficient solder-Minimum dimensions of solder bridge less than twice the diameter of the wire or less than 3/32 inch, whichever is greater. Entire area of contact between wire and terminal not jointed by solder bridge.	Improper wrap-more than l turn. Excess solder-Buildup solder on joint greater than necessary for good soldering, usually resulting in obliteration of wire contour.

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TABLE VII. Classification of defects for visual and mechanical inspection. continued.

Defect type	Classification	
	Major	Minor
Soldering	Cold solder joint-Chalky appearance, lacks metallic luster, presents rough "pileup" appearance, movement of wire or solder upon pick application. Rosin joint-Presence of excess rosin, relative movement of wire or solder upon pick application. Insulation in terminal hole - Solder over insulation, no appearances of visible wire contour.	Cold solder joint - Chalky appearance, lacks metallic luster, presents rough "pile-up" appearance; no relative action between wire and solder upon pick application
Wiring	Wiring not in accordance with schematic diagram. Broken strands more than 20 percent. In a 7-strand conductor, more than 2 broken strands. Insulation burned, abraded, pinched, or deteriorated between two or more conductors resulting in a potential short circuit. Taut wire-Wire exhibits no slack and subsequent breakage may occur due to stress on terminal parts. Insulation frayed to the extent that a potential short circuit exists.	Broken strands-20 percent or less. In a 7 strand conductor, 2 broken strands or less Insulation burned, abraded, pinched, or deteriorated, with exposure of bare wire, but short circuit not possible Taut wire-Slight stress on conductor but no possibility of subsequent breakage.

4.5.2 Electrical.

4.5.2.1 Input current. The dc input current to the microphone shall be measured as specified in the specification sheet.

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4.5.2.2 Output Impedance. The electrical output impedance of the microphone shall be tested as specified in the specification sheet.

4.5.2.3 Insulation resistance. The microphone shall be tested in accordance with MIL-STD-202, Method 302. There shall be no evidence of arcing or insulation breakdown during the test (see 3.8.1.3).

4.5.2.4 Dielectric withstanding voltage. The microphone shall be tested in accordance with MIL-STD-202, Method 301 (see 3.8.1.4). The following shall apply.

a. Test voltage 115 volts (V) rms.
b. Nature of potential - 60 Hz AC
c. Points of application - between one terminal and the external surface of the microphone assembly case.
d. There shall be no leakage current.
e. After removal of the test voltage, the microphone assembly shall meet the impedance specification at 1,000 Hz as specified in 4.5.2.2

4.5.3 Acoustical.

4.5.3.1 Sensitivity.

4.5.3.1.1 Sensitivity to a close sound source. Position the microphone under test with its face 0.25-inch on axis to the opening of an artificial voice (B&K type 4219 or equivalent) with the diaphragm of the microphone parallel to the driver unit. The artificial voice shall have been previously calibrated by means of a condenser microphone (B&K type 4133 or equivalent) certified traceable to the NIST (formerly the NBS). This calibration shall be conducted with an applied constant sound pressure level of 3.16 Pascal's (Pa) 104 dB sound pressure level, relative to 20 ?Pa. A regulated dc power supply shall be used to power the internal microphone amplifier. Determine the microphone output readings using an HP334A distortion analyzer or equivalent, set for voltage measurement (or a suitable electronic voltmeter having an input impedance of at least 0.5 megohm) connected as specified in the specification sheet.

4.5.3.1.2 Sensitivity at reduced bias voltage. As specified in the specification sheet (see 3.1). The sensitivity at reduced bias voltages shall be tested as specified in paragraph 4.5.3.1.1.

4.5.3.1.3 Sensitivity at altitude. The sensitivity to a close sound source shall be measured as specified (see 4.5.3 1.1) except that a simulated altitude of 15,000 feet, or as specified in the applicable specification sheet, shall prevail at the microphone.

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4.5.3.2 Frequency response.

4.5.3.2.1 Frequency response to a close sound source. The microphone shall be tested as specified in 4.5.3.1.1 except that a range of frequencies shall be used. The response of the microphone shall be tested in the frequency range of 400 to 6,000 Hz with a pure tone linear sweep from an audio oscillator (B&K type 1024 or equal) or at each l/3-octave frequency between 400 to 6, 000 Hz. The resultant plot of the "near field" frequency response of the microphone shall be compared to the limits as specified in the specification sheet.

4.5.3.2.2 Frequency response at altitude. The frequency response of the microphone to a close sound source shall be measured as specified (see 4.5.3.2.1) except that a simulated altitude of 15,000 feet, or as specified in the applicable specification sheet, shall prevail at the microphone.

4.5.3.3 Intelligibility. The microphones shall be subjected to a series of psychoacoustic tests using one of the three sets of speech materials specified in ANSI S3.2. The noise environment for the talkers shall be pink noise at 100 dB SPL presented in a diffuse field environment. The listeners shall be in a quiet room with less than 60 dB sound pressure level (SPL).

4.5.3.4 Noise immunity characteristic. The noise immunity characteristic of the microphone shall be determined by a comparison of the frequency response characteristic to a close sound source (near field) and the frequency response characteristic to a distant sound source in a “free field” (see 6.5.1). The following steps shall be performed to determine the noise immunity characteristic.

a. Plot the frequency response characteristic of the microphone to a close sound source (near field) as specified in 4.5.3.2.1.
b. Plot the frequency response characteristic of the microphone to a distant sound source (free field) as specified in 4.5.3.4.3.
c. Subtract the speaker correction factors (determined from the condenser microphone data in 4.5.3.4.3) from the 1/3 octave band (or narrow band) SPLs measured by the test microphone in step (b).
d. The resultant plot shall be the speaker-corrected free field response of the test microphone.
e. Subtract the near field response of step (a) from the free field response of step (d).

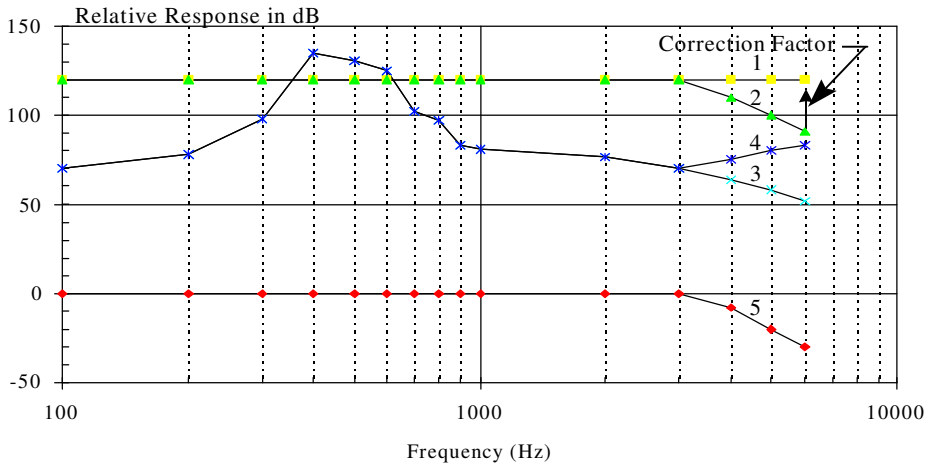
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4.5.3.4.1 Test conditions for measurement of a distant sound source (free field). Place the microphone assembly under test proximate to a condenser microphone (B&K type 4133 or equivalent) with calibration certified-traceable to the NIST (formerly the NBS) in the free field one meter from the sound source. (A suitable sound source can be constructed using an eight-inch chamber loudspeaker capable of generating 100 dB SPL of pure tone from 200 to 6,000 Hz at continuous output.) It is suggested that the speaker shall be housed in a speaker box with no dimension greater than 0.5 meter to ensure that the microphone is in the free field of the sound source.

4.5.3.4.2 Calibration of sound pressure level (SPL) of a distant sound source (free field). The sound source shall produce a sound pressure of 1 Pa (94 dB SPL, relative to 20 ?Pa) measured at 1,000 Hz by the condenser microphone.

4.5.3.4.3 Frequency response characteristic to a distant sound source (free field). The responses of the condenser microphone and the test microphone assembly shall be separately recorded in the range of 400 to 6,000 Hz with a pure tone linear sweep (narrow-band analysis) from an audio oscillator (B&K type 1024 plus power amplifier, if necessary) or at each 1/3-octave frequency between 400 and 6,000 Hz. The condenser microphone frequency response data shall be compared to the uncontrolled frequency response of the loudspeaker (see 4.5.3.4.1) at each 1/3-octave band (or narrow band) at the 104 dB SPL measured at 1,000 Hz. to determine "correction factors" (measured differences between the responses) for the loudspeaker response. These correction factors shall be used to normalize the loudspeaker frequency response to the condenser microphone as shown by the example plots on figure 1. If narrow-band analysis method is chosen, a computer can be used to normalize the loudspeaker to the required 94 dB SPL.

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Legend:

1. Manufacturer's specification for the condenser microphone.
2. Condenser microphone response plus chamber speaker response.
3. Test microphone assembly response plus chamber speaker response.
4. Normalized test microphone assembly response derived by use of correction factor 3 minus 5.
5. Normalized chamber speaker response derived by use of correction factor 2 minus 1

NOTE: Coordinate values for Relative Response and Frequency are shown for explanatory purposes only and do not represent actual values or scale.

FIGURE 1. Example plots showing method of normalizing loudspeaker frequency response between the responses for the loudspeaker response.

4.5.3.5 Linearity. The output voltage developed by the microphone across a specified resistive load as specified in the specification sheet (see 3.1) shall be measured at any one frequency within each of the following ranges: 400 to 800 Hz, 801 to 2,500 Hz, 2,501 to 4,000 Hz, and 4,001 to 6,000 Hz with input sound pressures of 74 dB, 84 dB, 94 dB, 104 dB, 114 dB, and 124 dB relative to 20 ?Pa.

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4.5.3.6 Harmonic distortion. The percent harmonic distortion in the output of the microphone assembly shall be determined by using a direct-reading harmonic distortion analyzer (Hewlett Packard 334A or equivalent) or by measuring the individual distortion products with a spectrum analyzer or a wave analyzer. The measurements shall be made at fundamental frequencies over the range of 200 to 6,000 Hz. with acoustic inputs having sound pressure levels (at the talking port) as specified in the specification sheet (see 3.1). If individual harmonic distortion products are measured, the total harmonic distortion (THD) shall be calculated using the following formula:

$$\text{Percent THD} = 100 \left[\text{the square root of } ? \left(A_i/F\right)^2\right]$$

Where A_i is the output voltage of each individual harmonic and F is the voltage of the fundamental frequency.

4.5.3.7 Gain stabilization time. While performing the sensitivity test to a close sound source as specified herein, the dc supply shall be disconnected for a period of at least 5 seconds and reconnected in sync with a timer for the purpose of recording gain stabilization time.

4.5.3.8 Overvoltage protection. The microphone assembly shall be subjected to an overvoltage condition as specified in the specification sheet

4.5.4 Cord, cordage, and cable.

4.5.4.1 Temperature range. A sample of unterminated cable, approximately 1.5 feet long, shall be subjected to the following test. The unterminated cable shall be placed in a test chamber at temperature of +85 ?1 ?C for a period of four hours. The cable shall be wrapped around a mandrel while exposed to this temperature without damage to the cable jacket or conductor insulation. The diameter of the mandrel shall be the same as the nominal cable diameter. The cable shall be attached to the horizontal mandrel and suspended vertically with the lower end weighted sufficiently, to keep the specimen taut and to permit wrapping around the mandrel without handling, by rotating the mandrel. The cable shall be inspected for damage or deformation and then placed in a test chamber at a temperature of -40 ?1 ?C for 20 hours. The cable shall then be wrapped about the mandrel in the same manner as it was for the high temperature test. The cable shall show no cracks or other damage to the jacket or conductor insulation.

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4.5.4.2 Flex life. Five samples of unterminated cable, each approximately 1.5 feet long, shall be subjected to the following test. Other flex test methods that will subject the cable to flex cycles equivalent to the flex cycles specified herein may be used upon approval by the contracting activity. The sample lengths shall be clamped and suspended through holes in a rectangular metal bar measuring 0.5 x 0.625 inches in the manner described below.

a. The cable shall be threaded through the 0.625-inch dimension of the bar. The diameter of the through holes shall be 0.005 to 0.010 inch greater than the diameter of the cable undergoing test.
b. The cable shall be clamped immediately above the metal bar such that the cable clamp shall rest on top of the metal bar and does not allow the cable to rotate in the bar. The through holes in the metal bar shall incorporate a radius of 0.125 inch at the end through which the cable emerges (bending point).
c. A weight of 0.5 pound shall be hung from the cable.
d. The two conductors plus the shield of each sample length of cable shall be connected in series to a 6V/150 ma lamp, and a suitable power supply to indicate electrical continuity.
e. The metal bar, supported horizontally, shall then be rotated about its longitudinal axis back and forth through an angle of 120 degrees (60 degrees each side of vertical) at a rate of approximately 85 cycles per minute. The arithmetic average of the number of flexes for each of the five sample lengths of cable before electrical discontinuity occurs shall be taken as the flex life of the cable.

4.5.4.3 Cord anchorage. With the connector firmly fixed, a pull of 8 pounds shall be applied to the cable in the direction of the axis of the microphone connector for a minimum of 30 minutes. The cord assembly shall be examined to determine whether any slippage has occurred at the connector clamp.

4.5.4.4 Conductor strain relief and twist. With the pins of one conductor anchored rigidly in the microphone and in a vertical position, a 5-pound weight shall be suspended from the pins of the other connector of the cord assembly for a period of at least 30 minutes. During the last 5 minutes of the test, the lower connector with the 5-pound weight suspended shall be twisted three turns clockwise and three turns counterclockwise from the normal resting position. The twisting cycle shall be repeated three times. The impedance of the microphone with cord assembly in place shall be measured at 1,000 Hz and a mechanical inspection shall be made before and immediately upon completion of the test. Upon completion of this test a dielectric withstanding voltage test shall be made on the cord assembly in accordance with the procedure specified in 4.5.4.5 after the impedance measurement and mechanical inspection have been made.

4.5.4.5 Dielectric withstanding voltage of the cord assembly. Each cord assembly shall be subjected to an alternative potential as outlined in 3.8.3.5. The voltage source shall have a capacity of not less than 3 kVA.

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4.5.4.6 Insulation resistance of the cord assembly. The insulation resistance of the finished cord assembly shall be measured in accordance with MIL-DTL-55668.

4.5.4.7 Pin retention test. The pin retention test shall be applied to all terminals and connectors of the amplifier and microphone unit. The pin to be tested shall be secured in a stationary holding device and the body holding the pin(s) shall be secured in a movable fixture attached to a force gage. The requirements of 3.8.3.7 shall be met.

4.5.4.8 Mounting screw torque/crush torque. The amplifier and microphone unit to be tested shall be installed on a mask and a torque of 3 inch pounds shall be applied. The requirements of 3.8.3.8 shall be met.

4.5.5 Environmental

4.5.5.1 Temperature-altitude. The microphone assembly shall be tested in accordance with MIL-STD-810, Method 520, Procedure III, except that humidity and vibration environments shall be deleted. Delete steps 2 through 7 and substitute the following. Microphones shall be checked for operation at room temperature between tests.

a. Step 2. With the microphone assembly nonoperating, adjust the chamber test conditions to -62?C and ambient pressure. The microphone temperature shall be stabilized and maintained for at least two hours.
b Step 3. With the microphone assembly nonoperating, adjust the chamber temperature to -54?C. The microphone shall be turned on and the altitude adjusted to 50,000 feet.
c. Step 4. With the microphone assembly nonoperating, adjust the chamber test conditions to -10?C and ambient pressure. After the microphone temperature has stabilized, the chamber door shall be opened and the frost permitted to form on the microphone. The door shall remain open long enough for the frost to melt, but not long enough to allow the moisture to evaporate.
d. Step 5. With the microphone assembly nonoperating, adjust the chamber test conditions to +85?C and ambient pressure. The chamber temperature shall be stabilized and maintained for at least 16 hours.

4.5.5.2 Humidity. The microphone assembly shall be tested in accordance MIL-STD-810, Method 507, Procedure III, except as follows.

a. Step 2. The chamber temperature shall be +71?C.
b. Step 4. Maintain 85 percent or greater relative humidity and reduce internal chamber temperature in 16 hours to 28 ? 10 ?C.
c. Step 5. Delete.
d. Step 6. Repeat steps 2, 3, and 4 for a total of 10 cycles (not less than 240 hours).
e. Step 7. The microphone assembly shall be operated within one hour following completion of the 10th cycle, and shall be tested to verify compliance with 3.8.2.1.1 and 3.8.2.2.1.

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4.5.5.3 Immersion. The microphone assembly shall be subjected to MIL-STD-810, Method 512, Procedure I. Following the immersion period, the microphone shall be tested to verify compliance with the requirements of 3.8.2.1.1 and 3.8.2.2.1.

4.5.5.4 Dust. The microphone assembly shall be subjected to MIL-STD-810, Method 510.3. Procedure I. The following conditions shall apply:

a. Step 2. The air velocity shall be 1,750 ? 250 feet per minute.
b. Step 5. The chamber temperature shall be +63 ?C and the air velocity shall be 300 ? 20 feet per minute.
c. Step 8. The equipment shall be tested to verify compliance with the requirements of 3.8.2.1.1 and 3.8.2.2.1.
d. Test duration shall be 6 hours at 23 ?C and 6 hours at 63 ?C.

4.5.5.5 Blast. Each microphone assembly being tested shall be mounted on the carriage of the U.S. Navy simulated gun blast equipment in accordance with Drawing NAWCADLKE-MISC-414100-0006. Placement shall be with the front edge of the microphone element in the test plane and with its axis coincident with that of the explosion chamber. The microphone assembly shall be subjected to 30 rounds of blast at a nominal peak pressure of 9.5 pounds per square inch. The equipment shall be tested to verify compliance with the requirements of 3.8.2.1.1 and 3.8.2.2.1.

4.5.5.6 Vibration. The microphone assembly shall be attached to the vibration table in a horizontal position, with the applied vibration in a direction perpendicular to the plane of the microphone. The microphone shall be subjected to sinusoidal vibration over a frequency range of 5 to 20 Hz at constant displacement double amplitude of 0.10 inch and over a frequency range of 20 to 500 Hz at a constant acceleration of 2G. The frequency of the applied vibration shall be swept logarithmically over the specified frequency range. One vibration cycle (5-500-5) shall consume 15 minutes. Total time of test shall be two hours. Repeat the above vibration in the fore/aft and side-to-side axis. The equipment shall be tested to verify compliance with the requirements of 3.8.2.1.1 and 3.8.2.2.1.

4.5.5.7 Shock-drop. The microphone assembly shall be dropped with random orientation twelve times onto a concrete surface from a height of 6 feet ?6 inches. The equipment shall be tested to verify compliance with the requirements of 3.8.2.1.1 and 3.8.2.2.1.

4.5.5.8 Electromagnetic environmental effects (E3) test requirements.

4.5.5.8.1 EMI tests. Compliance with the requirements of paragraph 3.8.4.8 and its subparagraphs shall be demonstrated by EMI tests. These EMI tests shall be performed in accordance with MIL-STD-461. For acquisition of data in support of these tests see 6.2.

4.5.5.8.2 EMI emission tests. Not applicable.

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4.5.5.8.3 EMI susceptibility tests. The requirements of 3.8.4.8.2.3 shall be tested in accordance with MIL-STD-461.

4.5.5.8.3.1 Conducted susceptibility tests. The microphone assembly power/signal cable conducted susceptibility test requirements shall be demonstrated using the detailed test techniques of MIL-STD-461. Applicable test methods for the microphone assembly are Test Method CS114 (Curve 5, high frequency limited to 30 MHz), Test Method CS115, and Test Method CS116 (Navy test levels).

4.5.5.8.3.2 Radiated susceptibility tests. The microphone assembly radiated susceptibility test requirements shall be tested in accordance with MIL-STD-461 and as follows. The radiated susceptibility tests shall be performed in accordance with Test Method RS103. The RS103 field levels shall be 200 V/m with scan rates as specified in MIL-STD-461. Modulation of the applied field shall be 1.05 kHz, 50 percent duty cycle, and pulse modulation. The detailed test set up to be used during the testing of the microphone assembly is detailed on figure 3. The test frequency range shall be limited from 2 MHz to 2 GHz.

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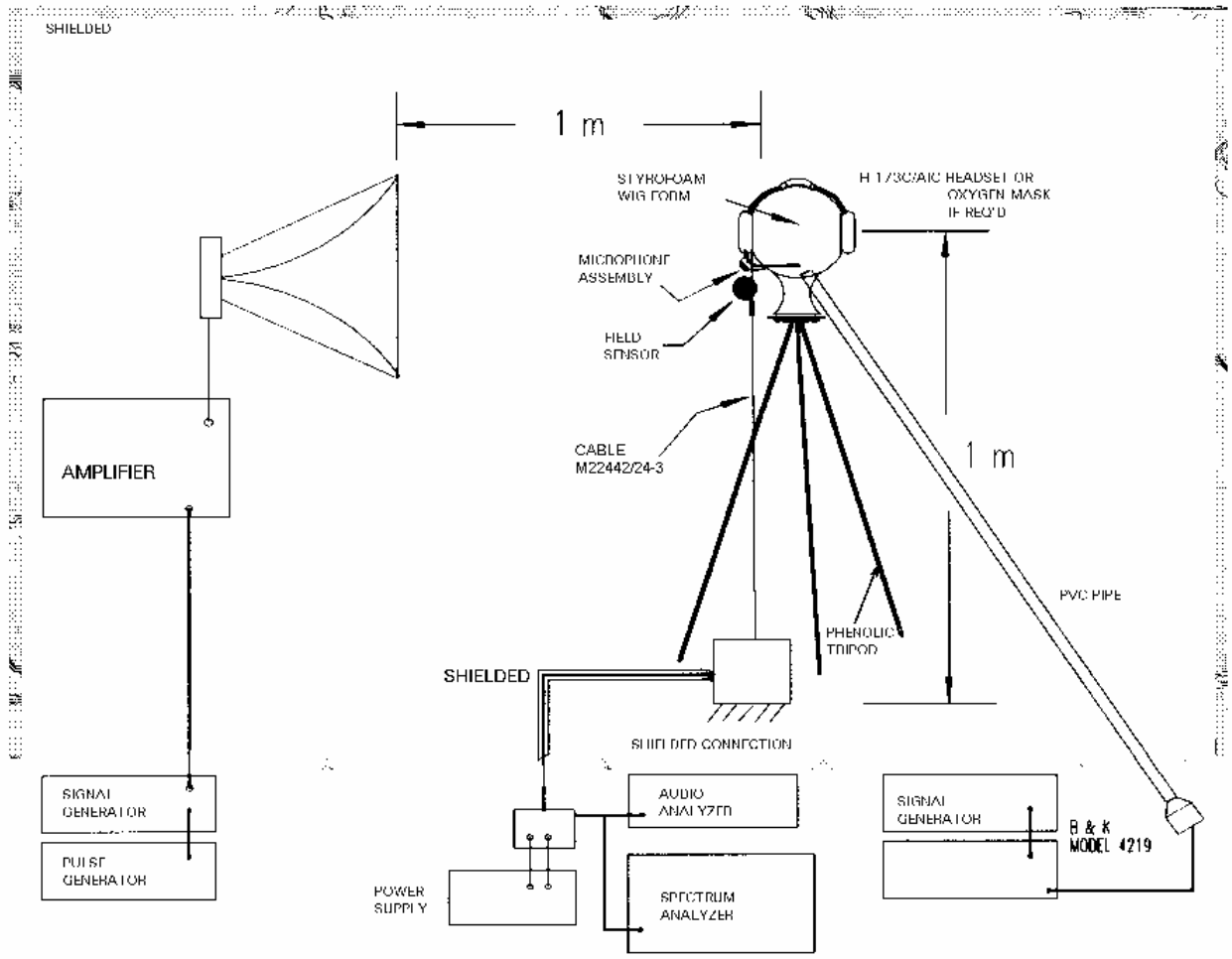


FIGURE 3. Radiated susceptibility test set up.

4.5.5.8.3.3 EMI lot sample tests. Lot samples of the microphone assembly shall meet MIL-STD-461 CS114 and RS103 test requirements when tested in accordance with this specification.

4.5.5.8.3.4 Intra-system EMC tests. Compliance to the requirements of 3.8.4.8.1 shall be verified by demonstration during aircraft system level testing. The test may be conducted by the government and shall be supported by the supplier.

4.5.5.8.3.5 Inter-system EMC tests. Compliance to the requirements of 3.8.4.8.1 shall be verified by demonstration during aircraft system level testing. The test may be conducted by the government and shall be supported by the supplier.

4.5.5.9 Explosive atmosphere when specified in the specification sheet (see 3.1). The microphone assembly shall be subjected to the explosive atmosphere test of MIL-STD-810, Method 511.3, Procedure I. The following conditions apply:

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a. Step 3. The microphone assembly shall be operated when specified using the dc bias voltages specified in the specification sheets. (See 3.1).
b. Step 4. The altitude shall be adjusted to the operating altitude of 50,000 feet. The equipment shall be tested to verify compliance with the requirements of 3.8.2.1.1 and 3.8.2.2.1.

4.5.5.10 Environmental stress screening (ESS). The contractor shall subject each unit to ESS when specified in the specification sheet.

4.5.5.11 Salt fog. The microphone assembly shall be salt fog tested in accordance with MIL-STD-810, Method 509, Procedure I. The following conditions apply:

a. Step 2. Following completion of the 48-hour exposure the microphone shall be removed from the test chamber and operated within two hours.
b. Step 3. The drying period shall be 72 hours.
c. Step 4. The microphone shall be tested to verify compliance with 3.8.2.1.1 and 3.8.2.2.1.

4.5.5.12 Fungus. The microphone assembly shall be tested for resistance to fungus growth in accordance with MIL-STD-810, Method 508.4. The units shall meet all performance requirements of this specification following this test. Compliance with this requirement may be demonstrated by analysis.

4.5.5.13 Mask pressurization. The microphone assembly shall be assembled on a mask. The portion of the microphone assemblies normally installed within the mask cavity shall be subjected to 60 cycles of pressurization. The exterior of the mask shall be at ambient pressure. Each of the 60 cycles shall consist of an internal pressure rise to 80 millimeters of mercury in one second, maintained for 10 seconds, and returned to ambient pressure and maintained for 48 seconds before starting the subsequent cycle. The microphone assembly shall be tested to verify compliance with 3.8.2.1.1 and 3.8.2.2.1.

4.5.5.14 Condensing humidity. The microphone assembly shall be assembled to a mask. The portion of the microphone assembly normally installed within the mask cavity shall be subjected to 10 cycles of humidity. Each cycle shall consist of 100 percent humidity at 45 ?C for 24 hours followed by <50 percent humidity at 25 ?C for 24 hours. The microphone assembly shall be tested to verify compliance with 3.8.2.1.1 and 3.8.2.2.1. The voltage potential of any area of the microphone assembly that could create a shock hazard by operator contact shall be continuously monitored with the microphone assembly under power and remain at zero volts (relative to system ground) throughout humidity testing.

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4.5.5.15. Rapid decompression. The microphone assembly shall be subjected to MIL-STD-810, Method 500.4, Procedure III. The following conditions shall apply:

a. Initial cabin altitude: 8,000 feet standard atmosphere pressure.
b. Final altitude: 40,000 feet after decompression and appropriate pressure for 10 minutes minimum.
c. Decompression rate: not more than 15 seconds.
d. Test item configuration: The microphone assembly shall be assembled to a compatible helmet assembly placed in the test chamber. Listeners in an external room shall monitor the performance of the microphone assembly before, during, and after the test. A continuous audio input shall be applied within 1/4-inch of the 'TALK' side of the microphone element.
e. Failure criteria: The microphone assembly shall recover without damage or degradation within 1.0 second of the decompression and meet the requirements of 3.8.2.1.1, 3.8.2.2.1 and 3.8.2.6 following the test.

4.5.5.16. Explosive decompression. The microphone assembly shall be subjected to MIL-STD-810, Method 500.4, Procedure IV. The following conditions shall apply:

a. Initial cabin altitude: 20,500 feet standard atmosphere pressure.
b. Final altitude: 50,000 feet after decompression and appropriate pressure for 10 minutes minimum.
c. Decompression rate: not more than 0.1 second.
d. Test item configuration: The microphone assembly shall be assembled to a compatible oxygen mask assembly placed in the test chamber. Listeners in an external room shall monitor the performance of the microphone assembly before, during, and after the test. A continuous audio input shall be applied within 1/4-inch of the 'TALK' side of the microphone element.
e. The test shall be repeated with an initial cabin altitude of 8,000 feet at one atmosphere and 25 ?C and final altitude of 23,000 feet and appropriate pressure for 10 minutes minimum. Decompression rate: not more than 0.1 second.
f. Failure criteria: The microphone assembly shall recover, without damage or degradation within 1.0 second of the decompression and meet the requirements of 3.8.2.1.1, 3.8.2.2.1 and 3.8.2.6 following the test.

4.5.5.17 Rain. The microphone assembly shall be subjected to the rain test as specified in MIL-STD-810, Method 506.3,Procedure I, Blowing Rain, for a period of one hour at a rate of four inches per hour. The microphone assembly shall be installed on a CBR mask. The mask shall be fitted on a dummy head and connected to a breathing simulator set to flow 3.9 ft^3/min at a rate of 44 cycles per minute. With the breathing simulator "ON" the mask shall meet the requirements specified in 3.8.4.17.

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4.5.5.18 Chemical agent permeation when specified in the specification sheet. Materials used to manufacture the microphone assembly shall be tested for chemical agent permeation. Test slabs made of the same formulation, thickness, cure time and cure temperature as the microphone assembly may be used in lieu of microphone assembly materials.

4.5.5.18.1 Mustard. Resistance to permeation by HD shall be verified by exposure to a liquid 10g/m^2 challenge for 16 hours with an endpoint of 4? g/cm^2 and a vapor challenge of 20? g/liter with an end-point of 0.16? g/cm^2 (vapor challenge in accordance with CRDC-SP-84010, Section 2).

4.5.5.18.2 Soman. Resistance to permeation by GD shall be verified by exposure to a liquid 10g/cm^2 challenge for 16 hours with an end-point of 1.25 ? g/cm^2 and a vapor challenge of 10? g/liter of GD for 16 hours with an end-point of 0.01? g/cm^2.

4.5.5.19 Quantitative fit factor (QFF). When specified in the specification sheet (see 3.1), the microphone assembly QFF shall be tested as follows. The QFF shall be determined in accordance with CRDC-SP-84010.

4.5.5.19.1 QFF Normal mode (chemical protection). A QFF of $2x10^4$ shall be measured and a test report generated using the methods of 4.5.5.19.

4.5.5.19.2 QFF Failure mode. A QFF of $1x10^3$ shall be measured and a test report generated using the methods of 4.5.5.19. Failure mode is defined as a non-operational blower unit.

4.5.6 Reliability. When required by the contract, the contractor shall perform detailed design reliability analysis. It is recommended that the part stress analysis method be in accordance with MIL-HDBK-217 for airborne inhabited transport. Derating of parts shall be performed in accordance with TE-00-AB-GTP-010. The results of the analysis shall be documented in accordance with the provisions of the contract.

5. PACKAGING.

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Services or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful. but is not mandatory.)

6.1 Intended use. These microphones are linear, noise-cancelling, dc powered high-gain microphones designed for use on military headband-type or helmet configured headsets at low altitudes or for use in pressure-type oxygen masks at altitudes where the use of an oxygen mask is required and CBR masks for special missions. The microphones are intended for use as a part of existing military and future airborne and ground communication systems that will provide communication of high intelligibility under the extreme noise conditions encountered in military environments.

6.2 Acquisition requirements. Acquisition documents should specify the following.

a. Title, number, and date of this specification
b. EMI test plan (see 4.5.5.8.1)
c. Reliability analysis if required (see 4.5.6)
d. Quantitative fit factor (see 4.5.5.19)
e. Packing required (see 5.1)
f. First article test plan (see 6.3)
g. First article test program (see 6.3).
h. If special or additional identification marking is required (see 6.4)

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6.3 First article. When first article inspection is required, the contracting officer shall provide specific guidance to offers whether the item(s) should be a preproduction sample, a first article sample, or a standard production item from the contractor's current inventory. The number of items to be tested shall be as specified in 4.3.l. The contracting officer shall also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Part or identifying numbers (PIN). Part numbers for microphones should be constructed as follows: M32015/X-Y with "X" being the specification sheet number and "Y" being any type of microphone on a specific specification sheet.

6.5 Definitions.

6.5.1 Free field A free field is a field in which the effects of the boundaries are negligible over the frequency range and distances of interest. For example, a free field exists in an anechoic chamber in which pressure level falls off at a rate of 6 dB per doubling of distance for the sound source of at least 200 Hz and a separation of at least l meter between the microphone and the sound source.

6.6 Subject term (key word) listing.

Amplifier
Electret
Frequency response
Harmonic distortion
High gain
Noise-canceling
Noise immunity
Sensitivity

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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